

Davide RAVOTTI

PERSONAL DATA

ADDRESS: Laboratoire Paul Painlevé, Université de Lille
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NATIONALITY: Italian

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SCIENTIFIC INTERESTS

My research focuses on the ergodic theory of smooth parabolic flows, including locally Hamiltonian flows on surfaces, unipotent flows on quotients of semisimple Lie groups, nil-flows on nilmanifolds, and their smooth perturbations. I am also interested in the following topics:

- Distributional limit theorems for dynamical systems.
- Infinite ergodic theory.
- Rigidity phenomena in homogeneous dynamics.

EDUCATION

2018	Ph.D. degree at the University of Bristol (Bristol, UK). Supervisor: prof. Corinna Ulcigrai. Thesis title <i>Mixing via shearing in some parabolic flows</i> . This thesis received a <i>Faculty of Science Commendation</i> .
2014	Diploma di Licenza of the Scuola Superiore of the University of Udine (Udine, Italy). Final mark: 110/110 <i>cum laude</i> .
2013	Master degree in Mathematics, University of Udine (Udine, Italy). Final mark: 110/110 <i>cum laude</i> .
2011	Bachelor degree in Mathematics, University of Udine (Udine, Italy). Final mark: 110/110 <i>cum laude</i> .

ACADEMIC POSITIONS

MAR. 2024-PRESENT	Chaire de Professeur Junior CNRS, Université de Lille (Lille, France).
JUN. 2021-FEB. 2024	Postdoctoral Assistant, University of Vienna (Vienna, Austria).
MAR. 2019-MAY. 2021	Research Fellow, Monash University (Melbourne, Australia).
SEP. 2018-FEB. 2019	Postdoctoral Assistant, University of Zürich (Zürich, Switzerland).
APR. 2018-AUG. 2018	Research Associate, University of Bristol (Bristol, UK).

FUNDED RESEARCH PROJECT AS PI

CPJ	2024-2029.
ESI RESEARCH IN TEAMS	2023 (together with H. Bruin, D. Terhesiu, and C. Fougerson).
AZRIELI FELLOWSHIP	2021-2023 (declined).

POSTDOCS

JÉRÔME CARRAND 2025-present.

PUBLICATIONS AND PREPRINTS

PREPRINTS:

- [18] D. Ravotti. Mixing asymptotics for time-changes of horocycle flows. Preprint, 2025 ([arXiv:2512.01488](#)).
- [17] A. Kanigowski, D. Ravotti. Multiple mixing for parabolic systems. Preprint, 2024 ([arXiv:2410.13686](#)).
- [16] R. Castorrini, D. Ravotti. Horocycle flows on abelian covers of surfaces of negative curvature. Preprint, 2024 ([arXiv:2405.08592](#)).

PUBLISHED/ACCEPTED:

- [15] H. Bruin, C. Fougerson, D. Ravotti, D. Terhesiu. On asymptotic expansions of ergodic integrals for $\mathbb{Z}^d$ -extensions of translation flows. *Studia Mathematica*, 284 (2025), 229–280.
- [14] P. Giulietti, A. Hammerlindl, D. Ravotti. Quantitative infinite mixing for non-compact skew products. *Discrete and Continuous Dynamical Systems*, 45 (2025), 3827–3856.
- [13] M. Artigiani, L. Flaminio, D. Ravotti. On rigidity properties of time-changes of unipotent flows. *Annali della Scuola Normale Superiore di Pisa. Classe di Scienze*, to appear.
- [12] E. Corso, D. Ravotti. Large hyperbolic circles. *Bollettino dell’Unione Matematica Italiana*, 18 (2025), 507–560.
- [11] L. Flaminio, D. Ravotti. Abelian covers of hyperbolic surfaces: equidistribution of spectra and infinite volume mixing asymptotics for horocycle flows. *Nonlinearity*, 37 (2024), Paper No. 075015.
- [10] D. Ravotti. Asymptotics and limit theorems for horocycle ergodic integrals à la Ratner, (with an appendix by Emilio Corso), *Journal de l’École polytechnique—Mathématiques*, Vol. 10 (2023), 305–334.
- [9] P. Giulietti, A. Hammerlindl, D. Ravotti. Quantitative global-local mixing for accessible skew products. *Annales Henri Poincaré*, 23 (2022), 923–971.
- [8] D. Ravotti. Polynomial mixing for time-changes of unipotent flows. *Annali della Scuola Normale Superiore di Pisa. Classe di Scienze*, Vol. XXIII (2022), 1491–1506.
- [7] A. Avila, G. Forni, D. Ravotti, C. Ulcigrai. Mixing for smooth time-changes of general nilflows. *Advances in Mathematics*, 385 (2021), Paper No. 107759.
- [6] D. Ravotti. Quantitative equidistribution of horocycle push-forwards of transverse arcs, *L’Enseignement Mathématique*, 66 (2020), 131–146.
- [5] A. Kanigowski, D. Ravotti. Polynomial 3-mixing for smooth time-changes of horocycle flows. *Discrete and Continuous Dynamical Systems*, 40 (2020), 5347–5371.
- [4] D. Ravotti. Parabolic perturbations of unipotent flows on compact quotients of $SL(3, \mathbb{R})$, *Communications in Mathematical Physics*, 371 (2019), 331–351.
- [3] D. Ravotti. Mixing for suspension flows over skew-translations and time-changes of quasi-abelian filiform nilflows, *Ergodic Theory and Dynamical Systems*, 39 (2019), 3407–3436.

- [2] D. Ravotti. Quantitative mixing for locally Hamiltonian flows with saddle loops on compact surfaces, *Annales Henri Poincaré*, 18 (2017), 3815–3861.
- [1] G. Panti, D. Ravotti. Measures induced by units. *Journal of Symbolic Logic*, 78 (2013), 886–910.

OTHER WRITINGS:

- [a] D. Ravotti. Introduction to Homogeneous Dynamics. Lecture notes for the course given at the University of Vienna, 2022.
- [b] D. Ravotti. Mixing via shearing in some parabolic flows. Ph.D. thesis (*Faculty of Science Commendation*), University of Brsitol, 2018.

TEACHING AND OTHER ACTIVITIES

STUDENT SUPERVISION:

UNIVERSITY OF VIENNA Bachelor Thesis of Gibeom Kim
Thesis title: *Discrete time Markov chains and applications*
Graduated in 2023.

LECTURER:

UNIVERSITÉ DE LILLE *Mathématiques élémentaires*, 2024.
 IMPAN SCHOOL *Ergodic and mixing properties of horocycle flows and their time-changes* (6h minicourse), May 2023.
 Lecture recording is available at <https://www.youtube.com/watch?v=IX-1wNIUwAc&t=3454s>
 UNIVERSITY OF VIENNA *Measure and Integration Theory*, 2023.
Ergodic Theory (together with H. Bruin), 2023.
Measure and Integration Theory (together with H. Bruin), 2022.
Homogeneous Dynamics, 2022.
 MONASH UNIVERSITY *Metric spaces, Banach spaces, Hilbert spaces* (together with J. Clutterbuck), 2020.
 ICTP SUMMER SCHOOL *Ergodic theory minicourse* (together with O. Butterley, I. Pasquinelli, S. Luzzatto, and L. Simonelli), July 2019.
 Lecture recording is available at <https://www.youtube.com/watch?v=PjFpuDiVL1U&t=632s>

SERVICE TO THE COMMUNITY

Referee for: *Inventiones Mathematicae*, *Crelle's Journal*, *Advances in Mathematics*, *Journal of Modern Dynamics*, *Ergodic Theory and Dynamical Systems*, *Monatshefte für Mathematik*, *Discrete and Continuous Dynamical Systems*.

LANGUAGE SKILLS

ITALIAN Native language
 ENGLISH full professional proficiency
 FRENCH intermediate proficiency
 SPANISH basic proficiency